

# Individual Project Report:

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## Introduction

This project is part of the development of an Automated Applicant Tracking System (ATS) Scorer, a tool designed to assess how well a candidate's resume matches a target job description.

As job applications become more competitive and automated resume screening becomes widespread, building a transparent and intelligent ATS scoring system is essential for both applicants and recruiters.

A key component of this system is the ability to process unstructured text data — from resumes and job descriptions — and extract structured, machine-readable information. This enables accurate scoring based on keywords, skills, competencies, and experience alignment.

## Project Objective

The primary objective of this module set is:

To automatically extract structured information from free-form resumes and job descriptions using advanced NLP techniques (LLM and rule-based), enabling fair and comprehensive scoring.

The extracted data will feed into the ATS Scorer pipeline, which evaluates alignment based on various dimensions such as:

- Keyword match
- Competency and skill relevance
- Experience depth and recency
- Overall ATS compatibility

## Components Developed

1 Job Description Extractor - LLM Based (jd\_llm\_extractor.py)

Purpose:

Utilizes a Large Language Model to extract detailed and nuanced information from job descriptions including required skills, qualifications, responsibilities, and soft skills.

Importance:

LLM-based extraction enables context-aware understanding, essential for competency analysis and skill gap identification.

## 2 Job Description Extractor - Rule-Based (JobDescriptionExtraction.py)

Purpose:

Provides a faster, fallback extraction method using rule-based or simpler NLP techniques.

Importance:

Ensures flexibility and efficiency, useful when LLM resources are unavailable or for quicker assessments.

## 3 Resume Extractor - LLM Based (resume\_llm\_extractor.py)

Purpose:

Processes resumes through an LLM to extract structured information such as contact information, skills, work experience, and education.

Importance:

Enables accurate keyword matching, experience relevance calculation, and ATS compliance checks.

## Integration Workflow

[Raw Job Description] → jd\_llm\_extractor.py OR JobDescriptionExtraction.py → [Structured JD Data]

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[ATS Scoring Engine]

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[Raw Resume] → resume\_llm\_extractor.py → [Structured Resume Data]

The ATS Scoring Engine then computes:

- Keyword Match Score
- Competency Match Score
- Experience Alignment Score
- Overall ATS Score

## Conclusion

The development of these modules successfully establishes the Information Extraction Layer of the ATS Scorer project.

With both rule-based and LLM-based extraction mechanisms, the system achieves:

- Context-aware understanding of job descriptions and resumes
- Operational flexibility and fallback capabilities
- Accurate and structured data representation for scoring

These capabilities make the ATS Scorer comparable to professional resume scanners, while providing actionable recommendations for users.